Open-book reference: Machines Ch1–9 (full course)

Cheat sheet · Use from Wed 11 Nov · pack REFM

Covers every chapter, Ch1 to Ch9, and replaces the short Ch1–3 strip of the earlier version. All phasors are **rms** and per phase unless marked “3ph”. Subscript 1 = phase (V_1, I_1), subscript L = line (V_L, I_L). Machines and transformers are **star-connected** unless the question says otherwise, and ratings are line V, line I and total (3ph) power. **p = pole-PAIRS** (course notation): $n_S = 60f/p = 120f/\text{poles}$. ϕ = angle of the current relative to the voltage, so lag means $\phi < 0$. Exam: 120 min, open book, 4 × 20 marks. Q1 = Ch1–2, Q2 = Ch3, Q3 = Ch7 with Ch4–5 basics in (a), Q4 = Ch8 and per-unit, with Q4c words only. Every check value below was recomputed in Python.

Constants: $\mu_0 = 4\pi \times 10^{-7}$ H/m · $g = 9.81$ m/s² · $\sigma_{Cu} = 59 \times 10^{6}$ (Ωm)⁻¹ · $\sqrt{2} = 1.414$ · $\sqrt{3} = 1.732$ · 1 kWh = 3.6 MJ · cm² = 10^{-4} m² · mm² = 10^{-6} m².

Ch1 Review of power and energy

Energy conversion (1.2). Transformer: AC electrical in, AC electrical out (changes V and I). Machine: electrical ↔ mechanical through magnetic fields. Motor: electrical in, mechanical out. Generator: the reverse. Machines are reversible. Gears and pulleys are “mechanical transformers”.

Item	Formula	Use / trap
Two key machine equations (1.2.3)	force $F = BIl$ (torque follows current); induced emf $E = Blv$ (voltage follows speed)	why $\tau = kI$ and $E = k\omega$ in Ch5
Fundamental limits (1.2.5)	current density 2–10 A/mm² (insulation temperature); flux density 1.5–2 T (saturation)	0.5 mm wire: area 0.196 mm², so about 0.4–2 A
Speed	$\omega = 2\pi n/60$ rad/s; $n = 60\omega/2\pi$ rpm	$n \approx 10\omega$ is a quick check. 1500 rpm = 157.1 rad/s
Mechanical power	$P_M = \tau\omega$; $\tau = Fr$; $F = mg$	rpm in $\tau\omega$ is wrong by ×9.55
rms and peak	$V_{pk} = \sqrt{2}V_{rms}$; $T = 1/f$	quoted mains values are rms. 240 V peak is 339.4 V; 120 V is 169.7 V
Time delay to angle	$\lvert\phi\rvert = \frac{\Delta t}{T} \times 360^\circ$	current peak **earlier** than voltage: leading, $\phi > 0$. 60 Hz: 90° = 4.167 ms
1ph power	$P = V_1 I_1 \cos\phi$, $Q = V_1 I_1 \sin(-\phi)$, $S = V_1 I_1$	pf = $\cos\phi$, always state lead or lag
Instantaneous 1ph power	$p(t) = V_1 I_1 \cos\phi + V_1 I_1 \cos(2\omega t + \phi)$	average = P; pulsates at 2f; goes negative when $Q \neq 0$
3ph power	$P = 3V_1 I_1 \cos\phi = \sqrt{3}V_L I_L \cos\phi$	balanced 3ph: total instantaneous power is **constant** (smooth torque, less vibration)
Efficiency	$\eta = P_{OUT}/P_{IN}$; $P_{LOSS} = P_{IN} - P_{OUT} = (1-\eta)P_{IN}$	4 unknowns (P_{IN}, P_{OUT}, P_{LOSS}, η), 2 equations: any two give the rest
From loss and η	$P_{IN} = P_{LOSS}/(1-\eta)$	1 kW loss at 90 %: $P_{IN} = 10$ kW,

Item	Formula	Use / trap
		$P_{OUT} = 9$ kW
Motor rating	rated power = **output** (shaft) power	$P_{IN} = P_{rated}/\eta$, never $P_{rated} \times \eta$
Energy cost	$E = Pt$ (kWh); cost = kWh × tariff	20 W, 4 h/day, 1 yr: 29.2 kWh, 11.68 dollars at 40 c/kWh

Why efficiency matters (1.7.4): (1) limited stored energy (EV range), (2) running cost (lifetime electricity cost is many times the purchase price), (3) cooling (the loss becomes heat, and temperature rise limits rating). IEC 60034-30-1 sets minimum efficiency classes; larger motors are more efficient.

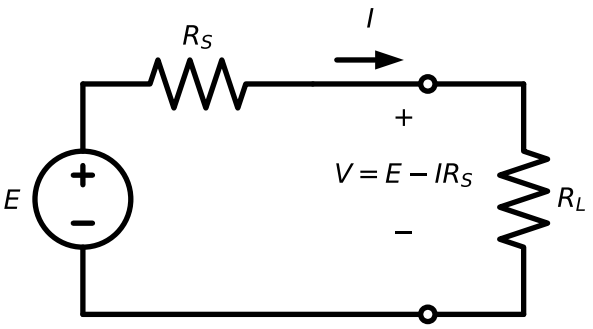
Power direction:

	Motor	Generator
P_{IN}	electrical $V_1 I_1 \cos\phi$ (×3)	mechanical $\tau\omega$
P_{OUT}	mechanical $\tau\omega$	electrical
pf from data	$\cos\phi = P_{IN}/(\sqrt{3}V_L I_L)$, lagging	$\cos\phi = P_{OUT}/(\sqrt{3}V_L I_L)$

CHECK VALUES: 2023 quiz Q1 (1.1 kW, 82 %, 1430 rpm): $P_{IN} = 1.341$ kW, loss 0.241 kW, ω = 149.7 rad/s, τ = 7.346 Nm. Jun24 Q1b (415 V, 6.8 A, 4 kW, 23 Nm, 1420 rpm): pf = 0.8184 lag, $P_{OUT} = 3.420$ kW, η = 85.50 %. PQ1.21 (110 V, 15 A, pf 0.8, 6 Nm, 1750 rpm): $P_{IN} = 1320$ W, $P_{OUT} = 1100$ W, η = 83.3 %. PQ1.22 (240 V phase, pf 0.85, 650 rpm, 63 Nm, 88 %): $P_{OUT} = 4.288$ kW, $P_{IN} = 4.873$ kW, $I_1 = 7.96$ A. PQ1.15 (240 V, 5 A): 54° lead gives pf 0.5878 lead, 705 W; 18° lag gives 0.9511 lag, 1141 W. Jun24 Q1a(ii): 1000 W in, 300 W loss: 700 W, 70 %; more loss lowers η, more input at the same loss raises η.

Ch2 Electric energy analysis

DC and battery model (2.1, 1.1). Ideal source E in series with R_S.



Battery model: E in series with R_S, terminal voltage V across the load

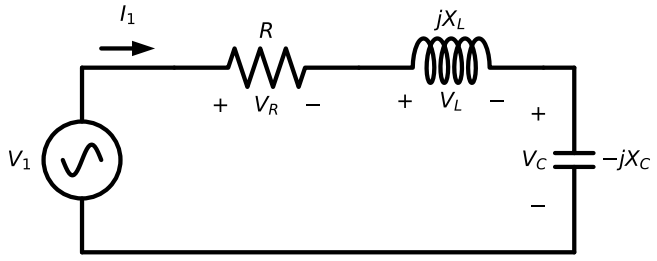
Item	Formula	Trap
Open circuit	$E = V_{OC}$ (no current, no drop)	
Internal R	$R_S = (E - V)/I = V_{OC}/I_{SC}$; two load points: $R_S = -\Delta V/\Delta I$	
Terminal V	$V = E - IR_S$	
Powers	load VI; loss I^2R_S; source $EI = VI + I^2R_S$	load power is VI, not EI
Max power	at $R_L = R_S$: $V = E/2$, $P_{max} = E^2/4R_S$, η = 50 %	never operated there
Resistance of a wire	$R = l/(\sigma A)$, $A = \pi d^2/4$	40 m of 0.5 mm Cu: 3.453 Ω

AC elements and phasors (2.2).

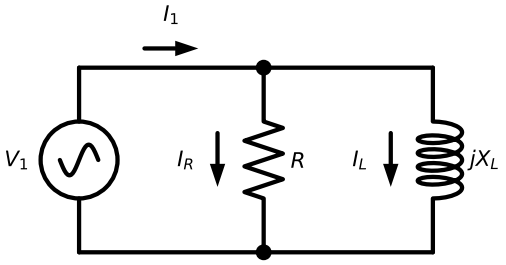
Element	Impedance	Current vs voltage	P	Q
R	R	in phase, $\phi = 0$	absorbs	0
L	$jX_L = j\omega L$	lags 90°	0	absorbs, $Q > 0$
C	$-jX_C = -j/(\omega C)$	leads 90°	0	generates, $Q < 0$

“CIVIL”: in a C, I leads V; V leads I in an L. Phasors rotate anticlockwise; “leading” = ahead of V. Multiply and divide in polar; add in rectangular. If Re < 0, add 180° to the calculator’s atan.

Series and parallel (2.3).



(a) series: same *I*, add *Z* (and voltage phasors)



(b) parallel: same *V*, add currents

(a) series RLC with element voltages; (b) parallel RL with branch currents

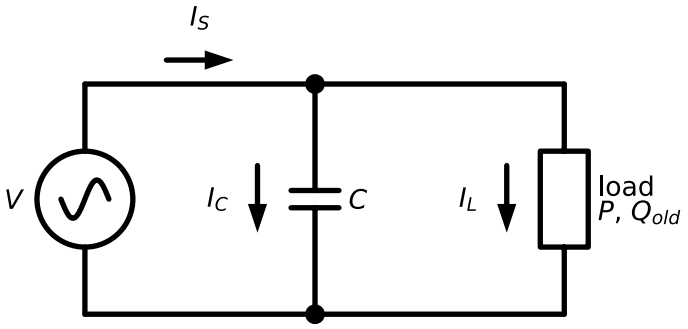
	Series RL (same I)	**Parallel RL (same V)**
Combine	$Z = Z_1 + Z_2$	$Z = Z_1 Z_2/(Z_1 + Z_2)$, or add branch currents
Magnitudes	$V_1^2 = V_R^2 + V_L^2$	$I_1^2 = I_R^2 + I_L^2$
P : Q	$R : X$	$X : R$
P, Q	I^2R, I^2X	V^2/R, V^2/X
R, X from P, Q	$R = P/I^2$, $X = Q/I^2$	$R = V^2/P$, $X = V^2/Q$
Phasor sketch	I is the reference	V is the reference

Series RLC: $|V_L|$ or $|V_C|$ can exceed V_1 (Jun23 1c: 650 V across C from 290.7 V). Resonance at $f_0 = 1/(2\pi\sqrt{LC})$, where $X_L = X_C$, Z = R and pf = 1. X is a signed scalar; Z is complex; $X < 0$ is capacitive. $L = X/\omega$, $C = 1/(\omega|X|)$.

Complex power and the power triangle (2.4).

Item	Formula	Note
Complex power	$S = V_1 I_1^* = P + jQ = S\angle -\phi$	conjugate the **current**. Forgetting it flips the sign of Q
Parts	$P = S\cos\phi$, $Q = S\sin(-\phi)$, $S = \sqrt{P^2+Q^2}$, pf $= P/S$	$\phi = -\arctan(Q/P)$
Sign of Q	$Q = \pm\sqrt{S^2 - P^2}$: sign from physics (L or lag +, C or lead −)	the root never gives the sign
Several loads	$P_{tot} = \sum P$, $Q_{tot} = \sum Q$, then S_{tot}	never add S's
Triangle	P horizontal, Q vertical (up = +), S hypotenuse, $-\phi$ at the origin	lagging: S above the P axis
Check	$P = \sum I^2 R$ (or $\sum V^2/R$), $Q = \sum I^2 X$ with signs	catches conjugate and angle errors

Power-factor correction (2.5). A capacitor in **parallel** with the load generates Q locally. P, and the load's own V, I, P, Q, are unchanged. The supply Q, S and current fall, and the supply pf rises towards 1. Over-correction past unity makes the pf leading and the current rise again, so one target pf has two C values (under- and over-corrected).



$I_S = I_L + I_C$; P unchanged, supply Q falls

PFC: capacitor C in parallel with the load across the supply

$$Q_S = \pm P\tan(\cos^{-1}\mathrm{pf}_{new})\ (+\ \text{lag},\ -\ \text{lead}), \qquad Q_C = Q_S - Q_{old} = -P(\tan\phi_1 - \tan\phi_2) < 0, \qquad C = \frac{|Q_C|}{\omega V^2}, \qquad X_C = \frac{V^2}{Q_C}, \qquad I_C = \omega C V$$

Unity pf: $Q_C = -Q_{load}$. Reverse problem (C given): $Q_C = -\omega C V^2$, $Q_S = Q_{old} + Q_C$, pf $= P/\sqrt{P^2 + Q_S^2}$, lead/lag from the sign of Q_S. Why it is useful: lower current, lower cable and transformer loading and I^2R loss, lower VA charges.

Equivalent-circuit parameter tests (2.6). Measure V, I, P; then $S = VI$, $Q = +\sqrt{S^2 - P^2}$ (inductive), then the series or parallel formulas above. The two models give different numbers for the same data: 240 V, 4 A, 500 W gives series $31.25 + j51.22\ \Omega$ but parallel $115.2\ \Omega \parallel j70.29\ \Omega$. 3ph: per phase first ($V_1 = V_L/\sqrt{3}$, $P_1 = P/3$). The transformer OC/SC tests (Ch3) and IM tests (Ch7) are this method.

Three-phase and star/delta (2.7–2.8).

(a) star and (b) delta loads on a three-phase supply

	Star (Y)	**Delta (Δ)**
Voltage	$V_L = \sqrt{3}V_1$, V_L leads V_1 by 30° (abc)	$V_L = V_1$
Current	$I_L = I_1$	$I_L = \sqrt{3}I_1$, I_L lags I_1 by 30°
Neutral	yes; balanced neutral current = 0	none
Power (both)	$S = 3V_1I_1 = \sqrt{3}V_LI_L$, $P = \sqrt{3}V_LI_L\cos\phi$, $Q = \sqrt{3}V_LI_L\sin(-\phi)$	same
Phase voltages	$V_a = V_1\angle 0^\circ$, $V_b = V_1\angle -120^\circ$, $V_c = V_1\angle +120^\circ$, sum = 0	

Conversion or comparison	Rule
Equivalent impedance	$Z_Y = Z_\Delta/3$ (then use star formulas); $Z_\Delta = 3Z_Y$
Same Z, same supply, star → delta	V_1 ×√3, I_1 ×√3, I_L ×3, P, Q, S ×3, pf unchanged
Same Z, delta → star	everything above ÷ (same factors); delta draws 3 × the star power
φ	always between phase V and phase I, never between V_L and I_L; 3ph pf = 1-phase pf
Standard route	$V_1 = V_L/\sqrt{3}$, $I_1 = I_L = S/(\sqrt{3}V_L)$, solve one phase, ×3 the powers
Capacitor bank, $\lvert Q_C\rvert/3$ each	star: $C_Y = \lvert Q_C\rvert/(\omega V_L^2)$ at $V_L/\sqrt{3}$; delta: $C_\Delta = \lvert Q_C\rvert/(3\omega V_L^2)$ at V_L; $C_Y = 3C_\Delta$
High-voltage transmission	$I_L = S/(\sqrt{3}V_L)$: 100 MVA is 209.9 A at 275 kV but 5249 A at 11 kV; loss $\propto I^2$

CHECK VALUES: Jun23 1a: E = 13.8 V, $R_S = 0.01433\ \Omega$, loss 1290 W, motor 2850 W, V(20 A) = 13.51 V ($P_{max} = 3322$ W, not asked). Jun23 1b (3 × 70 Ω, 460 V): star 3.794 A phase and line, 3.023 kW; delta 6.571 A phase, 11.38 A line, 9.069 kW. Jun23 1c ($I_1 = 13\angle 20°$; figure values 10 Ω, j30 Ω, –j50 Ω, inferred from the answers): $V_R = 130\angle 20°$, $V_L = 390\angle 110°$, $V_C = 650\angle -70°$ V, $Z = 22.36\angle -63.43°\ \Omega$, $V_1 = 290.7\angle -43.43°$ V, $I_1^* = 13\angle -20°$, $S = 1690 - j3380$ VA (sheet –j3379). Jun24 1c (100 Ω, j40 Ω at 2 A; values from the figure, inferred from the answers): R 400 W, 0 VAr; L 0 W, 160 VAr; $V_1 = \sqrt{50^2 + 20^2} = 53.85$ V. Jun24 1d: pf 0.8 lag; $P_S = 80$ W, $Q_S = 38.75$ VAr, $Q_C = -21.25$ VAr; $Q_C = -30$ VAr at 115 V, 60 Hz: C = 6.017 µF. 2023 quiz Q2: R = 49.31 Ω, X = 0.9028 Ω, inductive, L = 2.874 mH; $V_L = 1.5\angle -20°(20 + j5) = 30.93\angle -5.96°$ V. 2024 test Q1: 169.7 V, 120 V, 16.67 ms, 4.167 ms; 400 V, 230.9 V, 15 A, 15 A. Q2: $S = 1200 + j400 = 1265\angle 18.43°$ VA, pf 0.9487 lag. Q3 (240 V, 120 Ω ‖ j60 Ω): $I_R = 2\angle 0°$, $I_L = 4\angle -90°$, $I_1 = 4.472\angle -63.43°$ A, 480 W, 960 VAr. PQ2.17 street light (25 Ω ‖ j125 Ω, 240 V): $Z = 24.51\angle 11.31°$, $I = 9.79\angle -11.31°$ A, pf 0.9806 lag, $S = 2304 + j461$ VA. PQ2.18 (200 V, $V_R = 150$ V, 10 A): $V_L = 132.3$ V, pf 0.75 lag, R = 15 Ω, X = 13.23 Ω. PQ2.19 (parallel RC, 140 V, 500 W, 15 A): R = 39.2 Ω, Q = −2040 VAr, pf 0.238 lead, X = −9.61 Ω. PQ2.20 (100 Ω ‖ (10 + j20) Ω, 200 V, 60 Hz): $I = 10\angle -53.13°$ A, $S = 1200 + j1600$, pf 0.6 lag; to 0.8 lag: $Q_C = -700$ VAr, C = 46.4 µF, I falls 10 → 7.5 A. PQ2.29 (415 V, 15 kVA, 0.8 lag, star): $V_1 = 239.6$ V, $I_1 = 20.87\angle -36.87°$ A, 12 kW, 9 kVAr. PQ2.30 (delta 2 + j3 Ω, 415 V): $I_1 = 115.1$ A, $I_L = 199.4$ A, pf 0.555 lag, P = 79.5 kW, Q = 119.2 kVAr (in star: 26.5 kW, one third). PQ2.33 (415 V, 4.8 A, 2.5 kW): series 36.2 + j34.4 Ω; parallel 68.9 Ω ‖ j72.4 Ω. 3ph PFC example (20 kW, 415 V, 0.7 → 0.95 lag): $Q_C = -13.83$ kVAr, $C_Y = 255.6$ µF, $C_\Delta = 85.2$ µF, I_L 39.75 → 29.29 A.

Ch3 Electromagnetics and transformers

Magnetic circuits (3.1–3.5)

Gapped iron core with an N-turn coil and mean flux path

Magnetic equivalent circuit: mmf NI, core and gap reluctances in series

Electric	Magnetic	Units
current I	flux Φ	Wb
emf V	mmf $\mathcal{F} = NI$	At (A)
resistance $R = l/(\sigma A)$	reluctance $\mathcal{R} = l/(\mu_r \mu_0 A)$	At/Wb (= A/Wb)
field E	field strength $H = B/\mu$	A/m
current density J	flux density $B = \Phi/A$	T
conductivity σ	permeability $\mu = \mu_r \mu_0$	H/m

Formula	Use	Trap
$NI = \Phi \mathcal{R}_{tot} = \sum H_k l_k$; series add, parallel like resistors	flux, current, mmf	same Φ through every series section
$\mathcal{R}_g = l_g/(\mu_0 A)$, $\mathcal{R}_c = l_c/(\mu_r \mu_0 A)$	every Q2a	gap dominates: per metre it is μ_r times worse than iron
"Infinite permeability"	$\mathcal{R}_c = 0$, so $NI = B l_g/\mu_0$	
$L = N^2/\mathcal{R}_{tot} = \lambda/I$, $\lambda = N\Phi = LI$	inductance	gap lowers L and makes it linear
$X_M = \omega L$	AC winding reactance	
Toroid $l = \pi D_{mean}$; circular section $A = \pi d^2/4$	geometry	radius vs diameter
Gap count	saw cut or two poles "g apart" = **one** gap; two C-cores, or U-core plus armature = **two** gaps ($l_g = 2g$)	2024 test: paper 0.07 mm gives $l_g = 0.14$ mm
Fringing $A_g \approx (a + l_g)(b + l_g)$	only when asked	

Energy and force (3.4):

Formula	Use	Trap
$w = B^2/(2\mu_0)$ J/m³	energy density	
$E_{gap} = \frac{B^2}{2\mu_0} A l_g$; total $= \frac{1}{2} L I^2$	stored energy	l_g = **total** gap; iron term has an extra $1/\mu_r$
$F = \frac{B^2}{2\mu_0} A_{force}$	attraction	two pole faces: $A_{force} = 2A$
Hold a mass: $B = \sqrt{2\mu_0 mg/A_{force}}$	minimum B	
Constant current: $B \propto 1/l_g$, $F \propto 1/l_g^2$; constant B: F independent of gap	relays, actuators	

Saturation and leakage (3.5): above about 1.6–2 T (knee 1.2–1.7 T for silicon steel) μ_r collapses towards 1. Extra mmf buys almost no extra flux. With AC, V fixes the flux (sinusoidal), so the **current** becomes large and peaky and the winding overheats. This sets the voltage rating, and $F \propto I^2$ and constant L both fail. Leakage flux links one winding only, which gives X_1, X_2. Hysteresis loop: remanence B_r, coercivity H_c; loop area = hysteresis energy per m³ per cycle.

Losses (3.6)

Loss	Cause	Scaling	Model
Copper	I^2R in stator/rotor or primary/secondary windings	$\propto I^2 \propto$ (load)² ($\tau \propto I$)	series R
Hysteresis	energy to reverse magnetic domains	$P_H = k_H \hat{B}^2 f$	R_C
Eddy current	currents induced in the core	$P_E = k_E \hat{B}^2 f^2$	R_C
Friction (bearings)	constant torque loss	$\propto \omega$	in R_C for an IM
Windage (air, fan)	air drag	$\propto \omega^3$	in R_C for an IM

Reduce P_H: soft magnetic material (thin loop). Reduce P_E: thin insulated laminations and silicon steel (higher resistivity); eddy loss dominates at high f. No iron loss in air. Iron loss is set by V and f, not load: at fixed f both parts $\propto \hat{B}^2 \propto V^2$, hence $P_{FE} = V^2/R_C$. At fixed V with f rising, $\hat{B} \propto 1/f$, so $P_H \propto 1/f$ and P_E is constant.

Back-emf equation (3.7)

$$E_{rms} = 4.44\,N f \hat{\Phi} = 4.44\,N f \hat{B} A \qquad (4.44 = 2\pi/\sqrt{2};\ E \text{ rms},\ \hat{\Phi},\ \hat{B} \text{ peak})$$

Rearranged	Use	Trap
$N = V/(4.44 f \hat{B}_{max} A)$, round **up**	turns for a voltage	2023 quiz accepted 360 or 361
$\hat{B} = V/(4.44 N f A)$	actual flux density with integer N	
$V_{max} = 4.44 N f \hat{B}_{max} A$	voltage rating	
$N_2 = N_1 V_2 / V_1$ (nearest)	second winding; $E_1/E_2 = N_1/N_2$	same flux links both
New supply: $N \propto V/f$; same N: $V_{max} \propto f$	rewinding, 50 vs 60 Hz	halving f at fixed V doubles $\hat{B}$
E has **no** reluctance or μ_r in it	a cut gap changes the current, not the voltage	DC: current fixes mmf. AC: voltage fixes flux

Turns design (2023 quiz Q7): N sets $\hat{B}$ for a given V and f; choose it so $\hat{B}$ is just below saturation. Too few: $\hat{B}$ too high, saturation, large distorted current, overheating. Too many: core under-used, more winding R, more copper loss, lower η.

Ideal transformer and referral (3.7–3.8)

Ideal n:1 transformer with load Z_L

Item	Formula	Note
Turns ratio	$n = N_1/N_2 = V_1/V_2 = I_2/I_1$	n > 1 step-down; 240 → 24 V: n = 10 down; 240 → 480 V: n = 0.5 up
Ideal	$V_1 = nV_2$, $I_1 = I_2/n$, $S_1 = S_2$, $Z_{IN} = n^2 Z_L$	load looks n^2 bigger from a step-down primary
Refer secondary → primary	$V_2' = nV_2$, $I_2' = I_2/n$, $Z' = n^2 Z$, $R_2' = n^2 R_2$, $X_2' = n^2 X_2$	angles, P, Q and pf unchanged
Back to secondary	$V_2 = V_2'/n$, $I_2 = nI_2'$	to the LV side, ohms get **smaller**
Ratings	given as S (VA): V limited by saturation, I by copper loss	input VA = output VA (high η)
Rated currents	1ph $I = S/V$; 3ph $I = S/(\sqrt{3}V_L)$	16 kV/135 kV, 250 MVA: 9.021 kA, 1.069 kA
Ideal assumptions	no winding R, no leakage, infinite μ, no core loss	

Equivalent circuits (3.9–3.10)

Full transformer equivalent circuit referred to the primary

Simplified transformer equivalent circuit: R_C ‖ jX_M across V_1, then R + jX

	Full (referred to primary)	Simplified (used in exams)
Series	$R_1 + jX_1$ before the shunt, $R_2' + jX_2'$ after	$R = R_1 + R_2'$, $X = X_1 + X_2'$ in one series branch
Shunt	$R_C \parallel jX_M$ in the middle	moved to the input terminals across V_1
Why OK		shunt current ≲ 5 % of rated, so the error is small and the parts decouple

Element	Physical meaning
R_1, R_2	winding (copper) resistance: copper loss
X_1, X_2	leakage flux that links only one winding
X_M	magnetising mmf (finite permeability); current lags 90°
R_C	iron loss (hysteresis + eddy); current in phase

Tests (3.10.2)

OC test: meters on the LV side at rated voltage, HV side open

SC test: meters on the HV side at reduced voltage, LV side shorted

	Open-circuit (OC)	Short-circuit (SC)
Set-up	rated V on one side (usually LV), other side open	reduced V, about rated I, other side shorted (usually HV)
What is negligible	copper loss (tiny current): $P_{OC} = P_{FE}$	iron loss (tiny flux): $P_{SC} = P_{CU}$
Branch found	shunt (parallel RL)	series (series RL)
Formulas	$R_C = V^2/P$, $Q = \sqrt{S^2 - P^2}$, $X_M = V^2/Q$	$R = P/I^2$, $X = Q/I^2$ or $\sqrt{(V/I)^2 - R^2}$
3ph	per phase first: $V_L/\sqrt{3}$, $P/3$, I_L	same
Side	values come out on the side energised; $\times n^2$ to move LV → HV	
Reverse (params → readings)	$I = \sqrt{(V/R_C)^2 + (V/X_M)^2}$, $P = V^2/R_C$	$I = V/\lvert R + jX\rvert$, $P = I^2R$, $V_{SC} = I_{rated}\lvert R + jX\rvert$

Saturation makes X_M voltage-dependent (Prac 2: 3982 Ω at 100 V, 366 Ω at 240 V), so use the OC test at rated voltage.

Transformer on load, regulation and efficiency (3.8.8–3.10.8)

Item	Formula	Trap
Output conditions	$V_2' = V_{L1,rated}/\sqrt{3}\angle 0^\circ$ (3ph) or $V_{1,rated}\angle 0^\circ$ (1ph)	rated **primary** voltage (referred)
S	$S = P/\text{pf}$, or $\sqrt{P^2+Q^2}$ with pf $= P/S$, or "rated VA"	MW is not MVA
I_2'	$I_2' = \frac{S}{\sqrt{3}V_{L1}}\angle\phi$: $\phi = -\cos^{-1}$pf lag (absorbs Q), $+\cos^{-1}$pf lead (generates Q); or $\sin(-\phi) = Q/S$	1ph: S/V_2'
KVL	$V_1 = V_2' + I_2'(R+jX)$; $V_2' = V_1 - I_2'(R+jX)$	phasor sum in rectangular form
Divider	$V_2' = V_1\frac{Z_L'}{Z_L'+R+jX}$, with $Z_L' = n^2 Z_L$	
Input current	$I_1 = I_2' + V_1/R_C + V_1/(jX_M)$	
Phase of V_2'	resistive load: V_2' **lags** V_1 (series X makes I_2' lag V_1, and V_2' is in phase with I_2')	Jun23 2c(iii)
Line voltage	$V_{L1} = \sqrt{3}\lvert V_1\rvert$	
Regulation	$VR = \frac{V_{2NL}-V_{2FL}}{V_{2FL}} = \frac{\lvert V_1\rvert-\lvert V_2'\rvert}{\lvert V_2'\rvert}$	$V_{2NL}' = V_1$
Approximate drop	$\lvert V_1\rvert - \lvert V_2'\rvert \approx I_2'(R\cos\phi \pm X\sin\lvert\phi\rvert)$ (+ lag, – lead)	leading load can give VR < 0
Copper loss	$P_{CU} = 3I_2'^2R = x^2P_{CU,FL}$ (x = per-unit load)	use I_2', not I_1
Iron loss	$P_{FE} = 3\lvert V_1\rvert^2/R_C$, about constant	use V_1, not V_2'
Efficiency	$\eta = P_{OUT}/(P_{OUT}+P_{CU}+P_{FE})$, $P_{OUT} = S\,\text{pf}$; or $(P_{IN}-\text{losses})/P_{IN}$	×3 every loss for 3ph
Max η	at $P_{CU} = P_{FE}$: $x = \sqrt{P_{FE}/P_{CU,FL}}$; $\eta_{max} = \frac{xS\,\text{pf}}{xS\,\text{pf}+2P_{FE}}$	highest at unity pf; big transformers peak below rated load

Item	Formula	Trap
η vs load	0 at no load, rises, peaks where losses are equal, then falls	
OC scaling	V halved: I ×0.5, P ×0.25, Q ×0.25, pf ×1 (linear model)	a real core: I falls more (less saturation)
Ideal vs real, load R doubled	ideal: V unchanged, I halves; real: V rises slightly (less drop in R, X), I falls to slightly more than half	2024 test Q8

Three-phase transformers (3.8.9): treat as three single-phase units in star. Per phase: $V_1 = V_L/\sqrt{3}$, $I_2' = S/(\sqrt{3}V_{L1})$, solve the single-phase circuit, then ×3 all powers and losses and ×√3 for line voltages. "5 % above rated output voltage": $V_2' = 1.05V_{L1}/\sqrt{3}$ (16 kV: 9.699 kV).

Special transformers (3.11): isolation (1:1, unearthed supply, medical); voltage (potential) transformer, e.g. 11 kV:110 V; current transformer, e.g. 10 kA:5 A with a 1-turn primary needs 2000 secondary turns; autotransformer (Variac): one winding with a moving wiper, no isolation, 0 to about 280 V. RCD (safety switch): trips when active and neutral currents differ by tens of mA; it does not protect against active-to-neutral contact or a fault with no earth path. High-frequency model and electronic transformers: not examinable.

CHECK VALUES: Jun23 2a (20 cm², 50 cm, N = 32, μr = 3000, 1.5 mm): $\mathcal{R}_g = 596.8$ kA/Wb, $\mathcal{R}_c = 66.31$ kA/Wb, L = 1.544 mH; E_{gap}(1.8 T) = 3.867 J; loss 12 W (30/70) at 1.5 T, 60 Hz → 1.2 T, 100 Hz: $P_H = 3.84$ W, $P_E = 14.93$ W, total 18.77 W. Jun23 2b (240 V:32 V; SC 27 V, 9.7 A, 55 W; OC 240 V, 1.3 A, 45 W): R = 0.5845 Ω, X = 2.721 Ω, $R_C = 1280$ Ω, $X_M = 186.6$ Ω. Jun24 2a (d = 2.5 cm, N = 120, 0.5 mm, infinite μ): A = 4.909 cm², $\mathcal{R}_g = 810.6$ kA/Wb, L = 17.77 mH, V_{max}(1.5 T, 50 Hz) = 19.62 V, F = 439.5 N. Jun24 2b (6:1, 12 V; figure values inferred from the answers: 0.3 Ω load, 15 Ω series): $Z_1 = 0.3$, $Z_2 = 10.8$, $Z_3 = 25.8\ \Omega$; $I_1 = 0.4651$ A, $I_2 = 2.791$ A, $V_1 = 5.023$ V, $V_2 = 0.8372$ V. Jun24 2c (16 kV/220 kV; R = 0.14, X = 0.28, $R_C = 250$, $X_M = 14\ \Omega$): $Z_{OC} = 13.98\angle 86.79^\circ\ \Omega$, $I_1 = 660.9$ A, P = 1.024 MW; ×0.5, ×0.25, ×0.25, ×1. 2023 quiz Q3 (D = 80 mm, 20 cm², μr = 1000, 1.5 T, 240 V, 50 Hz): N = 360.4 → 360/361; $\mathcal{R} = 100$ kA/Wb, L(200 turns) = 0.4 H, X = 125.7 Ω, I(100 V) = 0.7958 A (sheet 0.7955 from X = 125.6). Q4 (4 mm, 20 cm², 75 turns, 1.1 T): I = 46.69 A, F = 962.9 N. Q5: $P_{CU} = 1.179$ kW, $P_{FE} = 1.369$ kW, η = 97.78 % (sheet 97.79). Q6: $V_2' = 1.905\angle 0^\circ$ kV, $I_2' = 6.056\angle -38.74^\circ$ A; (b) $V_1 = 2.254\angle 5.89^\circ$ kV, line **3.904 kV** (sheet 3.883 kV does not match its own phase value). 2024 test Q4: 0.3519 T, 0.1086 T, $B_{min} = 0.0860$ T. Q5: $N_1 = 168.9$ → 169, R = 3.453 Ω, $N_2 = 33.8$ → 34. Q6: R = **0.3068** Ω (sheet 0.3086, digit swap), X = 1.005 Ω. Q7 (14 kV, 150 MVA, generating 35 MVAr): $V_2' = 8.083\angle 0^\circ$ kV, $I_2' = 6.186\angle +13.49^\circ$ kA. Lecture 3.10.5 (2.2 kV/220 V, 100 kVA, OC 2.2 kV, 2.5 A, 1 kW; SC 150 V, 45.5 A, 2.15 kW; rated, 0.82 lag): $R_C = 4840$, $X_M = 510.9$, R = 0.3462, X = 1.872 Ω; $V_1 = 1306\angle 1.54^\circ$ V, line 2.262 kV, VR = 2.84 %, $P_{CU} = 715.2$ W, $P_{FE} = 1057.5$ W, η = 97.88 % (slide 97.90 % from a mistyped 4893 Ω); max η at x = 1.18. PQ3.29 (R = 0.609, X = 2.776, $R_C = 1280$ Ω; 2400 VA, 0.87 lag, n = 10): $|V_1| = 259.9$ V, VR = 8.27 %, $P_{CU} = 60.9$ W, $P_{FE} = 52.8$ W, η = 94.8 %. PQ3.30 (3 kVA, 415 V/48 V, R = 0.95, X = 8.5 Ω, $R_C = 3300$ Ω, unity pf): $V_1 = 246.1$ V, line 426.3 V, 49.7 W, 55.1 W, η = 96.6 %. PQ3.27 reverse (240 V, $R_C = 3200$, $X_M = 550$, R = 1.1, X = 3.5 Ω): OC 0.443 A, 18.0 W; SC at 240 V 65.4 A, 4.71 kW; rated 4.17 A needs 15.3 V. Toroid PQ3.5–3.8 (15 cm, μr = 1000, d = 2 cm, N = 100, 1 T): I = 3.75 A, L = 8.378 mH; with a 1 mm gap I = 11.71 A, L = 2.683 mH. PQ3.9 (2 mm, 20 cm², 100 turns, 12 A): B = 0.754 T, E = 0.905 J, F = 452 N, L = 12.57 mH. PQ3.11 (40 cm, 10 cm², μr = 5000, 150 turns, 1.6 T): L = 0.353 H, $I_{max} = 0.679$ A, $E_{max} = 53.3$ V.

Ch4 Machine basics

SYMBOLS: f supply frequency (Hz); n speed (rpm); n_S synchronous speed (rpm); n_{NL} DC no-load speed (rpm); n_R, τ_R, P_R rated speed, torque (Nm), output power (W); $x = \tau/\tau_R$ torque fraction.

Formula	Use	Trap
$n_S = \frac{60f}{p} = \frac{120f}{\text{poles}}$	every AC question	8 poles means p = 4, so 750 rpm at 50 Hz
$p = 60f/n_R$: integer means synchronous; non-integer means induction, take $p = \lfloor 60f/n_R \rfloor$	"how many poles / which type"	round **down** for an induction machine ($n_S > n_R$)
Ideal η: IM $= n_R/n_S = 1 - s$; DC motor $= n_R/n_{NL} = E/V$; DC gen $= V/E$	"ideal (maximum possible) efficiency"	counts rotor (or armature) copper loss only
Linear τ–n: $n = n_0 - x(n_0 - n_R)$, with $n_0 = n_{NL}$ (DC) or n_S (IM). Sync: $n = n_S$	speed at part torque	
$P/P_R = x \cdot n/n_R$ (sync: $= x$)	part-load output power	**not** just x: the speed rises

50 Hz synchronous speeds: 2p 3000 · 4p 1500 · 6p 1000 · 8p 750 · 10p 600 · 12p 500 rpm (60 Hz: ×1.2).

Speed control (Jun24 Q3a):

Machine	τ–n shape	To halve speed safely	What stays constant
DC	sloped line from n_{NL}	halve V	speed drop $n_{NL} - n_R$ (curve slides parallel)
Induction	steeper line from n_S	halve V **and** f (V/f constant)	slip speed $n_S - n$ at a given torque
Synchronous	vertical line at n_S	halve V **and** f	—

Changing f alone raises $\hat{\Phi} \propto V/f$, which saturates the core and causes overcurrent.

Machine types: DC for toys, cars, battery tools. Universal (series) for small mains appliances (drills, blenders). Induction for everything else (pumps, fans, lifts). Synchronous for power generation and servo drives. Constant-torque loads (conveyors, cranes) need high starting torque. Fan loads have $\tau \propto \omega^2$, $P \propto \omega^3$.

Ch5 DC machines

SYMBOLS: E back-emf (V); k machine constant (V/(rad/s) = Nm/A); I armature current (A), positive **into** the machine; R armature resistance (Ω); V terminal voltage (V).

DC machine equivalent circuit (motor convention)

Formula	Use	Trap
$E = k\omega$	speed from emf, or the reverse	ω in **rad/s**. For 220 V at 1500 rpm, $k = 220/157.1 = 1.40$
$\tau = kI$	torque from current	same k as for E
$V = E + IR$ (motor convention)	KVL	generator: $I < 0$, or $E = V + \lvert I \rvert R$
$P_{IN} = VI$; $P_{OUT} = EI = \tau\omega$; $P_{loss} = I^2R$	power bookkeeping	input is VI, not EI
No-load: $I = 0$, $E = V$, $\omega_{NL} = V/k$	find $k = V/\omega_{NL}$	
Stall: $\omega = 0$, $E = 0$, $I_S = V/R$, $\tau_S = kI_S$	find $R = V/I_S$	$I_S \gg I_R$, so it overheats
$P_{M,max}$ at $n_{NL}/2$ and $\tau_S/2$, with η = 50 %		

Sign conventions:

	Motoring	Generating
Speed	$n < n_{NL}$	$n > n_{NL}$
emf	$E < V$	$E > V$
Current, torque	$I > 0, \tau > 0$	$I < 0, \tau < 0$
Ideal η	E/V	V/E

A wind generator charging a battery motors when the wind drops ($E < V$). A series diode blocks the reverse current. Commutator: rectifies the coil emf so the output is DC. Wound-field

machines (not examinable): $k \propto I_F$. A shunt machine has $\omega \approx V/k$. A series (universal) machine has $\tau \propto I^2$ and runs on AC or DC.

Ch6 Rotating fields

- Each phase is a pair of opposite coils. The three magnetic axes are 120° apart. Positive current gives a field along the axis and negative current reverses it. This gives 6 field directions, which step round to make a **stepwise** rotating field (stepper motor, not examinable).
- **Balanced 3ph currents** give a field of **constant magnitude** rotating at **constant speed** n_S.
- Phase sequence sets the direction: abc = bca = cab one way, acb = cba = bac the other. **Swapping any two supply leads reverses rotation.** A cyclic relabelling of all three does not.
- One electrical cycle moves the field one pole-pair, which is $1/p$ of a revolution. Hence $n_S = 60f/p$.
- A single-phase winding gives a **pulsating** field (forward plus backward rotating), so the starting torque is zero. It needs an auxiliary winding and capacitor to start (not examinable).

Ch7 Induction machines

SYMBOLS: s slip (dimensionless); f_2 rotor frequency (Hz); $\omega_S = 2\pi n_S/60$ synchronous speed (rad/s); $\omega = (1-s)\omega_S$ rotor speed (rad/s); R_1 stator resistance (Ω); R_2' rotor resistance referred to the stator (Ω); $X = X_1 + X_2'$ total leakage reactance (Ω); R_C core-loss resistance (Ω); X_M magnetising reactance (Ω); I_2' referred rotor current (A); I_M magnetising-branch current (A); P_{AG} airgap power (W, 3ph).

Induction machine per-phase simplified circuit: R_C ‖ jX_M across V_1, then R_1, jX and R_2′/s in series

Simplified circuit (the course uses this for every calculation): $R_C \parallel jX_M$ sits across V_1 at the terminals. The series branch is $R_1 + jX + R_2'/s$. R_1 and R_2'/s are **not** lumped. Compared with a transformer: the secondary rotates and is shorted, the rotor frequency is sf, the rotor R is divided by s, and the airgap makes X_M small (magnetising current 25–60 % of rated, against about 5–10 %).

Formula	Use	Trap
$s = \frac{n_S - n}{n_S}$; $n = (1-s)n_S$; $f_2 = sf$	first line of every Q3b/c	use the **operating** speed. Keep 4 s.f.
$\frac{R_2'}{s} = R_2' + R_2'\frac{1-s}{s}$	heat plus mechanical	$s = 0$ gives ∞ (open); $s < 0$ gives a negative resistance
$I_2' = \frac{V_1}{(R_1 + R_2'/s) + jX}$	rotor current	$V_1 = V_L/\sqrt{3}$ for star
$I_M = \frac{V_1}{R_C} - j\frac{V_1}{X_M}$; $I_1 = I_2' + I_M$	input current	add in **rectangular** form
$P_{IN} = 3V_1 I_1 \cos\phi$	input power	$\phi = \angle I_1$ with $V_1 \angle 0^\circ$
Stator Cu $= 3I_2'^2 R_1$; iron $= 3V_1^2/R_C$	losses	R_1 carries I_2', not I_1
$P_{AG} = 3I_2'^2 R_2'/s = P_{IN} - \text{Cu}_1 - \text{Fe}$	airgap power	uses R_2'/s
Rotor Cu $= 3I_2'^2 R_2' = sP_{AG}$; $P_M = (1-s)P_{AG}$	split $s : (1-s)$	rotor Cu is **not** divided by s
$\tau = \frac{P_{AG}}{\omega_S} = \frac{P_M}{\omega}$	torque	P_{AG} pairs with ω_S
$\tau = \frac{3V_1^2\,(R_2'/s)}{\omega_S[(R_1 + R_2'/s)^2 + X^2]}$	torque only (“use the torque formula”)	phase V_1; $\tau \propto V_1^2$ at a fixed s
$s_{\tau\max} = \frac{R_2'}{\sqrt{R_1^2 + X^2}}$; $n_{\tau\max} = (1 - s_{\tau\max})n_S$	pull-out point	magnitude $\lvert R_1 + jX\rvert$, not $R_1 + X$
$\tau_{max} = \frac{3V_1^2}{2\omega_S\left(R_1 + \sqrt{R_1^2 + X^2}\right)}$	check value	independent of R_2'
$\tau_{ST} = \tau(s{=}1)$; $I_{ST} \approx V_1/\lvert R_1 + R_2' + jX\rvert$	starting	$I_{ST} \approx$ 5–6 × rated
$R_{2EXT}' = \sqrt{R_1^2 + X^2} - R_2'$	τ_{max} at standstill (wound rotor)	“total rotor R” = $\lvert R_1 + jX\rvert$; “external” = minus R_2'
V/f: scale V_1, X and ω_S by f/f_R; R_1, R_2' unchanged	inverter drive	τ_{max} is about constant, and drops slightly at low f

Typical values: rated slip 0.01–0.09 (rated f_2 about 0.5–5 Hz). $\tau_{max} \approx 2\tau_R$. No-load current 25–60 % of rated, at low pf. Ideal η = $1 - s$. Rotors: squirrel cage (aluminium bars and end rings, fixed R, cheap) or wound rotor (slip rings, external R raises $s_{\tau\max}$ and starting torque but not τ_{max}, at the cost of rotor loss).

Sign conventions:

	Motor ($0 < s \leq 1, n < n_S$)	**Generator ($s < 0, n > n_S$)**
R_2'/s, P_{AG}, τ, P_{IN}	all positive	**all negative**
$\angle I_1$	about −10° to −45°	about −145° to −180°
Q_{IN}	positive	**still positive** (needs magnetising VArs)
Power order	$P_{IN} > P_{AG} > P_M$	shaft in $= \lvert P_{AG}\rvert(1-s) > \lvert P_{AG}\rvert >$ electrical out
Efficiency	P_M/P_{IN}	$\lvert P_{IN}\rvert/\lvert\tau\omega\rvert$

Speed for a given load torque τ_L. Quick: $s \approx s_R\,\tau_L/\tau_R$. Exact: let $x = R_2'/s$, then solve $\tau_L\omega_S x^2 + (2R_1\tau_L\omega_S - 3V_1^2)x + \tau_L\omega_S(R_1^2 + X^2) = 0$. Take the **larger** x (the smaller slip).

Ch8 Synchronous machines and per-unit

SYMBOLS: S_B base power, 3ph total (VA); V_B base voltage, **line** (V); I_B base current (A); Z_B base impedance (Ω); E_0 excitation (open-circuit) voltage (V or pu); δ rotor angle = $\angle E_0$ relative to V (°); X_S synchronous reactance (Ω or pu); R stator resistance (Ω); I_F field current (A).

Synchronous machine per-phase model: E_0∠δ behind R + jX_S, terminal voltage V∠0° (generator convention)

Formula	Use	Trap
$I_B = \frac{S_B}{\sqrt{3}V_B}$, $Z_B = \frac{V_B^2}{S_B}$, $P_B = Q_B = S_B$	bases (1ph: $I_B = S_B/V_B$)	Z_B uses line V and **total** VA
pu = actual / base. In pu: $S = VI^*$, $P = VI\cos\phi$	all Ch8 work	no √3 or 3 in pu. Line pu = phase pu
$\lvert Z_S\rvert = \frac{V_{OC,L}/\sqrt{3}}{I_{SC}}$, $X_S = \sqrt{\lvert Z_S\rvert^2 - R^2}$	X_S from OC/SC tests	both readings at the **same** I_F; √3 on V_{OC}
$E_0 = V + I(R + jX_S)$	generator convention, I **out**	$V = V\angle 0°$
$E_0 = V - I_m(R + jX_S)$	motor, I_m **into** the machine	gives the same E_0, with $\delta < 0$
$I = \frac{S}{V}\angle\phi$: $\phi = -\cos^{-1}$ pf lag, $+\cos^{-1}$ pf lead; or $I = (S/V)^*$	current phasor	use the **actual** V (e.g. 0.97)
$P = \frac{VE_0}{X_S}\sin\delta$ pu (3ph)	power angle	SI phase V: ×3; SI line V: already 3ph
$P_{max} = VE_0/X_S$ at $\delta = 90°$	stability limit	keep the **old** E_0. Practical limit about 75°
$Q_{gen} = \frac{VE_0\cos\delta - V^2}{X_S}$ pu	check only (derived)	+ means it delivers Q
$I_F = E_0(\text{pu}) \times I_F(E_0 = 1\text{ pu})$; $I_F(1\text{ pu}) = I_F^* V_B/V_{OC}$	field current	linear (air-gap) line only
$\lvert Q\rvert_{max} = \sqrt{S_{rated}^2 - P_{E'}^2}$, $P_E = P_M/\eta$	capability, PFC	use electrical P_E, not P_M

Sign conventions (lead/lag, over/under-excited):

Case	Current	Excitation	Q	Phasor
Generator, lagging output	I behind V	over-excited	delivers Q	E_0 long, δ small
Generator, leading output	I ahead of V	under-excited	absorbs Q	E_0 short, δ large, low P_{max}
Motor, leading input	I_m ahead of V	over-excited	supplies Q (this is PFC)	$\delta < 0$
Motor, lagging input	I_m behind V	under-excited	absorbs Q	$\delta < 0$

Rule: **over-excited generates Q, under-excited absorbs Q**, in either mode. $\delta > 0$ is generating, $\delta < 0$ is motoring. The mechanical input (steam valve) sets P and δ. The field current I_F sets E_0, and so Q.

V-curve (constant P): $|I|$ against I_F is a V, with its minimum at pf = 1. A larger P moves the V up and to the right. Say “generates Q / absorbs Q” rather than lead or lag. Slide 8.6.4 labels pf from the supply side.

Construction facts: round rotor for 2 or 4 poles (thermal plants); salient pole for 4 poles and up (hydro, low speed). PM rotors are mostly used in motors, wound-field rotors in generators. Efficiency is up to 98 %. To **synchronise**, match the phase sequence, frequency, voltage magnitude and phase. Close the contactor when the lamps are dark. Not self-starting on a fixed-frequency supply (average torque zero at standstill).

Ch9 Power systems introduction

Part	Rating	Notes
1. Generation	3ph synchronous generators, 10–25 kV	coal, gas, nuclear, hydro, wind, solar; often near the energy source
2. Transmission	3ph, 115–800 kV (SA: 275 kV, 132 kV)	high V for efficiency: $I = S/(\sqrt{3}V_L)$, loss I^2R
3. Distribution	3ph, 2.4–69 kV (SA: 66, 33, 11 kV)	within the city
4. Loads	domestic 1ph 240 V (safety); commercial/industrial 3ph 415 V to several kV	industrial loads mostly induction motors
5. Market	National Electricity Market (AEMO)	

Transmission system (9.1.3): drawn as a **single-line diagram** (one line represents 3ph). A **busbar** is a node where every point is at the same (nominal) voltage. Generators and lines are **paralleled** for reliability (one can fail), maintenance without interrupting supply, and running the most efficient number of generators for the demand.

Scale numbers: household about 2 kW; SA demand typically about 3000 MW; largest coal, nuclear, hydro stations about 1000 MW; large wind turbine 2–5 MW each. Torrens Island: 1963, gas, 8 turbines, 1280 MW. Pelican Point: 2000, gas, 2 turbines, 485 MW, 53 % efficient. Demand peaks in the evening; off-peak water heating fills the night.

Generation (9.2): via mechanical power: thermal (coal, gas, nuclear, geothermal, biomass, solar thermal → steam → turbine), water (hydro, tidal, wave) or wind. Direct: solar cells (silicon, tens of kW up to tens of MW) and fuel cells (hydrogen + oxygen → electricity; electrolysis is the reverse). Conventional plants: reliable, cheap to run, controllable, but limited fuel (gas tens of years, coal hundreds) and waste (CO_2 and global warming; sulphur and nitrogen oxides and acid rain; nuclear waste). Renewables: “unlimited” resource and zero emissions in operation, but difficulties are **economic** (cost, government support), **technical** (variability of wind, surviving storms) and **environmental** (visual impact, noise).

PQ9.1 answers: (a) advantages of electric power: made from many sources, transmitted long distances, easily converted to other forms, many applications. (b) sources: oil, gas, coal, nuclear, hydro, tidal, wave, wind, solar, geothermal. (c) fossil-fuel issues: finite resources, greenhouse gas emissions and pollution. (d) high voltage: for a fixed power the current is inversely proportional to voltage and the loss goes with current squared, so loss falls (doubling V quarters the loss). (e) lines are paralleled for reliability and maintenance.

Procedures

P1–P7 are Q1 (Ch1–2), P8–P16 Q2 (Ch3), P17–P23 Q3 (Ch4–7), P24–P28 Q4 (Ch8). Each ends with a check value.

P1. Motor power, torque, efficiency, pf (Q1b)

1. $\omega = 2\pi n/60$. Write it down (a mark).
2. $P_{OUT} = \tau\omega$ for a motor (the rating is output). $P_{IN} = P_{OUT}/\eta$; loss $= P_{IN} - P_{OUT}$.
3. Electrical side: 1ph $P = V_1 I_1 \cos\phi$; 3ph $P = \sqrt{3} V_L I_L \cos\phi$. So pf $= P/(\sqrt{3} V_L I_L)$, "lag" for a motor.
4. Torque from power: $\tau = P/\omega$.

Check: Jun24 Q1b pf 0.8184 lag, 3.420 kW, 85.50 %. 2023 quiz Q1 τ = 7.346 Nm.

P2. Battery or DC source (Q1a)

1. Draw E in series with R_S. $E = V_{OC}$.
2. $R_S = (E - V)/I$ from the loaded reading.
3. Loss $I^2 R_S$, load VI, check $EI = VI + I^2 R_S$.
4. New load: $V = E - IR_S$.

Check: Jun23 1a 0.01433 Ω, 1290 W, 2850 W, 13.51 V.

P3. Star or delta load (Q1b, Jun23)

1. The supply voltage is the line voltage. Star: $V_1 = V_L/\sqrt{3}$. Delta: $V_1 = V_L$.
2. $I_1 = V_1/|Z|$. Star: $I_L = I_1$. Delta: $I_L = \sqrt{3} I_1$.
3. $P = 3I_1^2 R$ (or $3V_1 I_1 \cos\phi$). Cross-check $\sqrt{3} V_L I_L \cos\phi$.
4. Delta = 3 × star power for the same Z.

Check: Jun23 1b star 3.794 A, 3.023 kW; delta 6.571 A, 11.38 A, 9.069 kW.

P4. Phasor circuit and complex power (Q1c)

1. Every element as an impedance: R, $+jX_L$, $-jX_C$ (from ωL, $1/\omega C$ if needed).
2. Reference: the supply V at 0°, unless a phasor is given (Jun23: $I_1 = 13\angle 20^\circ$).
3. Series: $V_k = IZ_k$, $Z = \sum Z_k$, $V = IZ$. Parallel: $I_k = V/Z_k$, $I = \sum I_k$.
4. Add in rectangular, convert to polar at the end.
5. $S = VI^*$: write I^* explicitly (a mark). P = Re, Q = Im, pf with lead/lag.
6. Magnitudes only: series $V = \sqrt{V_R^2 + V_L^2}$, parallel $I = \sqrt{I_R^2 + I_L^2}$.
7. Element powers: R: $I^2 R$, 0 VAr; L: 0 W, $I^2 X$.
8. Phasor sketch if asked: label every phasor, show right angles.

Check: Jun23 1c $V_1 = 290.7\angle -43.43^\circ$ V, $S = 1690 - j3380$ VA. 2024 test Q3 $I_1 = 4.472\angle -63.43^\circ$ A.

P5. R and X from measurements (2023 quiz Q2, PQ2.33)

1. 3ph: per phase first.
2. Q given, or $Q = \pm\sqrt{(VI)^2 - P^2}$ with the sign from the physics.
3. Series: $R = P/I^2$, $X = Q/I^2$ (or $|Z| = V/I$, $X = \sqrt{|Z|^2 - R^2}$). Parallel: $R = V^2/P$, $X = V^2/Q$.
4. Q > 0 inductive: $L = X/\omega$. Q < 0 capacitive: $C = 1/(\omega|X|)$. Say why ("absorbs Q").

Check: 2023 quiz Q2 R = 49.31 Ω, X = 0.9028 Ω, L = 2.874 mH.

P6. Power-factor correction (Q1d)

1. Load P and Q_{old} (from P, Q, or S and pf, or I^2R, I^2X). Sum over loads. Sketch the triangle; pf $= P/S$.
2. P is unchanged: $P_S = P$.
3. $Q_S = P\tan(\cos^{-1}\mathrm{pf}_{new})$, + for lag, − for lead, 0 for unity.
4. $Q_C = Q_S - Q_{old}$ (negative: "absorbs −21.25 VAr").
5. $C = |Q_C|/(\omega V^2)$ with V across C and the stated f (60 Hz in Jun24).
6. 3ph bank: $|Q_C|/3$ per capacitor at $V_L/\sqrt{3}$ (star) or V_L (delta). State the connection.
7. If asked: $S_{new} = P/\mathrm{pf}_{new}$, $I_{new} = S_{new}/V$; draw both triangles on the same P.

Check: Jun24 1d 38.75 VAr, −21.25 VAr, 6.017 µF. PQ2.20 46.4 µF.

P7. Fill-in-the-blanks (Q1a, 2024 test Q1)

1. rms/peak: ×√2. Period $1/f$; angle → time: $\Delta t = (\phi/360^\circ)T$.
2. Star: $V_1 = V_L/\sqrt{3}$, $I_1 = I_L$. Delta: $V_1 = V_L$, $I_1 = I_L/\sqrt{3}$.
3. Turns ratio $n = V_1/V_2$; n > 1 step down, n < 1 step up.
4. Efficiency: $P_{OUT} = P_{IN} - P_{LOSS}$; more loss → η down; more input at the same loss → η up.

P8. Reluctance and inductance (Q2a)

1. Units to m and m². Toroid $l = \pi D$; circular $A = \pi d^2/4$.
2. Count the gaps (Ch3 "Gap count" row).
3. $\mathcal{R}_g = l_g/(\mu_0 A)$, $\mathcal{R}_c = l_c/(\mu_r\mu_0 A)$ (0 if infinite µ). Add.
4. $L = N^2/\mathcal{R}_{tot}$. AC: $X = 2\pi f L$, $I = V/X$.
5. Target B: $\Phi = BA$, $NI = \Phi\mathcal{R}_{tot}$. Given I: $\Phi = NI/\mathcal{R}_{tot}$, $B = \Phi/A$.

Check: Jun23 2a 596.8 and 66.31 kA/Wb, 1.544 mH. Jun24 2a 810.6 kA/Wb, 17.77 mH.

P9. Energy, force, DC current for a B (Q2a, 2023 quiz Q4, 2024 test Q4)

1. DC current: infinite µ: $NI = Bl_g/\mu_0$. Finite: $NI = B(l_c/\mu_r\mu_0 + l_g/\mu_0)$.
2. B from a current: $B = NI/(l_c/\mu_r\mu_0 + l_g/\mu_0)$.
3. Gap energy $\frac{B^2}{2\mu_0}Al_g$ (total gap length).
4. Force $\frac{B^2}{2\mu_0}A_{force}$ ($2A$ for two faces). Hold a mass: $F = mg$, solve B.

Check: Jun23 2a(ii) 3.867 J. Jun24 2a(iii) 439.5 N. 2023 quiz Q4 46.69 A, 962.9 N. 2024 test Q4 0.3519, 0.1086, 0.0860 T.

P10. Back-emf: turns, V_{max}, B (Q2a, 2023 quiz Q3, 2024 test Q5)

1. $E = 4.44Nf\hat{B}A$ (rms E, peak B), $V \approx E$.
2. Turns: $N = V/(4.44f\hat{B}_{max}A)$, round **up**. Actual $\hat{B}$ with the integer.
3. $V_{max} = 4.44Nf\hat{B}_{max}A$.
4. Secondary: $N_2 = N_1V_2/V_1$, nearest integer. Wire: $R = l/(\sigma\pi d^2/4)$.
5. New supply: $N \propto V/f$.

Check: 2023 quiz Q3 360.4 → 360/361. 2024 test Q5 169, 3.453 Ω, 34. Jun24 2a(ii) 19.62 V.

P11. Iron-loss scaling (Q2a(iii))

1. Split the total into P_{H1} and P_{E1}.
2. New $\hat{B}$: given, or $\hat{B} \propto V/f$.
3. $P_{H2} = P_{H1}(B_2/B_1)^2(f_2/f_1)$; $P_{E2} = P_{E1}(B_2/B_1)^2(f_2/f_1)^2$.
4. State both parts and the total. Inverse ("what B for the same loss"): solve the combined factor for B^2, then square-root.

Check: Jun23 2a(iii) 3.84 W, 14.93 W, 18.77 W.

P12. Ideal transformer referral (Q2b, Jun24)

1. $n = N_1/N_2$ ("6:1" is n = 6).
2. Refer the secondary impedances: $Z' = n^2Z$. Name each "Z seen" the question labels.
3. Solve the primary loop: $I_1 = V_S/Z_{tot}$, $V_1 = I_1Z'_L$.
4. Back: $I_2 = nI_1$, $V_2 = V_1/n$. Check $V_1I_1 = V_2I_2$.

Check: Jun24 2b 0.3, 10.8, 25.8 Ω; 0.4651 A, 2.791 A, 5.023 V, 0.8372 V.

P13. OC and SC tests → parameters (Q2b, 2024 test Q6)

1. 3ph: $V_1 = V_L/\sqrt{3}$, $I_1 = I_L$, $P_1 = P/3$.
2. SC (low V) → series: $R = P_1/I_1^2$, $Q_1 = \sqrt{(V_1I_1)^2 - P_1^2}$, $X = Q_1/I_1^2$. Check $\sqrt{R^2 + X^2} = V_1/I_1$.
3. OC (rated V) → shunt: $R_C = V_1^2/P_1$, $Q_1 = \sqrt{(V_1I_1)^2 - P_1^2}$, $X_M = V_1^2/Q_1$.
4. Say which side; $\times n^2$ to move to the other side.
5. Draw the simplified circuit with values as impedances ("$j2.721\ \Omega$").

Check: Jun23 2b 0.5845, 2.721, 1280, 186.6 Ω. 2024 test Q6 0.3068, 1.005 Ω.

P14. Transformer on load: V_2', I_2', V_1, regulation (Q2c, 2023 quiz Q6, 2024 test Q7)

1. Simplified circuit per phase, referred to the primary, labelled with impedances.
2. $V_2' = V_{L1,rated}/\sqrt{3}\angle 0°$ (1ph: $V_{1,rated}$).
3. S from P/pf, $\sqrt{P^2 + Q^2}$ or the rating. $|I_2'| = S/(\sqrt{3}V_{L1})$.
4. Angle: lag $-\cos^{-1}$pf; lead or "generating Q" $+$; from Q: $\sin(-\phi) = Q/S$.
5. $V_1 = V_2' + I_2'(R + jX)$ in rectangular form, then polar. Line $= \sqrt{3}|V_1|$.

6. Regulation $(|V_1| - |V_2'|)/|V_2'|$. Lagging: a few % positive, V_1 leads V_2' by a few degrees.
7. "Expression only" (Jun23 2c): KVL $V_2' = V_1 - I_2'(R_1 + R_2' + jX_1 + jX_2')$; divider $V_2' = V_1 R_L'/(R_L' + R_1 + R_2' + jX_1 + jX_2')$; phase: lags.

Check: 2023 quiz Q6 6.056∠−38.74° A; 2.254∠5.89° kV, 3.904 kV line. 2024 test Q7 6.186∠+13.49° kA. Lecture 3.10.5 2.262 kV, 2.84 %.

P15. Losses, efficiency and maximum efficiency (2023 quiz Q5)

1. $P_{CU} = 3I_2'^2 R$ (I_2', not I_1).
2. $P_{FE} = 3|V_1|^2/R_C$ (V_1, not V_2').
3. $P_{OUT} = S\,\text{pf}$, or $P_{IN} - P_{CU} - P_{FE}$ if P_{IN} is given.
4. $\eta = P_{OUT}/(P_{OUT} + P_{CU} + P_{FE})$ or $(P_{IN} - \text{losses})/P_{IN}$.
5. Max η: P_{FE} = OC power at rated V; $P_{CU,FL}$ = SC power at rated current (or $3I_{rated}^2 R$). $x = \sqrt{P_{FE}/P_{CU,FL}}$, load $= xS_{rated}$, $\eta_{max} = xS\,\text{pf}/(xS\,\text{pf} + 2P_{FE})$; state the pf.

Check: 2023 quiz Q5 1.179 kW, 1.369 kW, 97.78 %. Lecture 3.10.5 97.88 %, x = 1.18.

P16. Open circuit and scaling (Jun24 2c)

1. $V_1 = V_L/\sqrt{3}$. Only the shunt branch conducts: $Z = 1/(1/R_C + 1/jX_M)$.
2. $I_1 = V_1/|Z|$ (or $\sqrt{(V_1/R_C)^2 + (V_1/X_M)^2}$). $P = 3V_1^2/R_C$, $Q = 3V_1^2/X_M$.
3. V halved: I ×0.5, P ×0.25, Q ×0.25, pf ×1.

Check: Jun24 2c 660.9 A, 1.024 MW.

P17. Poles, type, ideal η, linear τ–n (Q3a)

1. $p = 60f/n_R$. Integer: synchronous. Not integer: induction, round down. Poles = 2p. $n_S = 60f/p$.
2. Ideal η: IM n_R/n_S; DC motor n_R/n_{NL}.
3. Part torque x: $n = n_0 - x(n_0 - n_R)$; $P/P_R = x\,n/n_R$.

Check: Jun23 3a 10 poles, induction, 93.33 %, 590 rpm, 26.34 %.

P18. New AC speed at torque fraction x (speed control)

1. $\Delta n_R = n_0 - n_R$ (rated speed drop).
2. New zero-torque speed: $n_{0,new} = n_{target} + x\,\Delta n_R$.
3. $f_{new} = f_R\,n_{0,new}/n_0$, then $V_{new} = V_R\,f_{new}/f_R$. (DC: $V_{new} = V_R\,n_{NL,new}/n_{NL}$.)
4. Sketch: DC and IM lines slide parallel; synchronous vertical line moves. Halving speed safely: DC halve V; AC halve V and f.

P19. DC operating point (Ch5)

1. Get k (from no-load, E/ω or τ/I) and R (from the stall test).
2. Chain the unknowns: $\tau \leftrightarrow I$ by k; $I \leftrightarrow E$ by $V = E + IR$; $E \leftrightarrow \omega$ by k.
3. Convert ω to rpm last.
4. Find P_{IN}, P_{OUT}, η. Check that η equals E/V.

5. State motor or generator from the sign of I, or from n against n_{NL}.

P20. IM no-load and blocked-rotor tests → parameters (Q3b)

1. Convert to per phase: $V_1 = V_L/\sqrt{3}$ and $P_1 = P/3$. $I_1 = I_L$ for star.
2. DC test: R_1 per phase. Halve a line-to-line reading in star.
3. Blocked rotor ($s = 1$, ignore R_C and X_M): $R = P_1/I_1^2 = R_1 + R_2'$, so $R_2' = R - R_1$.
4. $Q_1 = \sqrt{(V_1 I_1)^2 - P_1^2}$, then $X = Q_1/I_1^2$. If a split is needed, $X_1 = X_2' = X/2$.
5. No load ($s \approx 0$, ignore R_1 and X_1): $R_C = V_1^2/P_1$. $Q_1 = \sqrt{(V_1 I_1)^2 - P_1^2}$, then $X_M = V_1^2/Q_1$. (R_C also absorbs friction and windage.)
6. Draw the circuit with every value written as an impedance.

Check: Jun24 3b $R_1 = 2$, $R_2' = 3.2$, X = 7.635 Ω.

P21. IM equivalent circuit → τ (Q3c full solve)

1. Header block: $V_1 \angle 0^\circ$; p from the rated speed; n_S; ω_S; s at the operating speed; R_2'/s; ω.
2. State motor or generator from the sign of s.
3. $Z = (R_1 + R_2'/s) + jX$, in both rectangular and polar form.
4. $I_2' = V_1/Z$.
5. If the shunt branch is kept: $I_M = V_1/R_C - jV_1/X_M$, then $I_1 = I_2' + I_M$ in rectangular form. Otherwise $I_1 = I_2'$.
6. pf $= \cos \angle I_1$ (say lag). $P_{IN} = 3V_1 I_1 \cos\phi$.
7. $P_{AG} = 3I_2'^2 R_2'/s$, then $\tau = P_{AG}/\omega_S$. Check it with the torque formula.
8. $P_M = \tau\omega = (1-s)P_{AG}$, then $\eta = P_M/P_{IN}$.
9. Check: $P_{IN} - 3I_2'^2(R_1 + R_2') - 3V_1^2/R_C = P_M$.

Check: Jun23 3c τ = 101.7 Nm, $I_1 = 29.79 \angle -38.44^\circ$ A. Jun24 3c $I_1 = 2.029 \angle -174.2^\circ$ A, −1451 W, −1481 W (sheet −1.482 kW), −14.15 Nm.

P22. Maximum torque and R_{EXT} (Q3b)

1. $|R_1 + jX| = \sqrt{R_1^2 + X^2}$.
2. $s_{\tau\max} = R_2'/|R_1 + jX|$, then $n_{\tau\max} = (1 - s_{\tau\max})n_S$.
3. τ_{max} = the torque formula at $s_{\tau\max}$ (check with the closed form).
4. For τ_{max} at standstill: $R_{2EXT}' = |R_1 + jX| - R_2'$.
5. Sketch: the peak height stays the same, and the peak moves to lower speed as rotor R rises. Starting torque rises up to τ_{max} (0.5 pu and 1 pu of R_{EXT} in Jun23 3b(ii)).

Check: Jun23 3b $R_{2EXT}' = 2.328\ \Omega$.

P23. Speed for a given load torque: see the end of Ch7 (quick $s \approx s_R T_L/\tau_R$; exact quadratic, larger root in R_2'/s).

P24. Per-unit conversion (Q4)

1. S_B = machine rating (3ph VA). V_B = rated line voltage.

2. $I_B = S_B/(\sqrt{3}V_B)$ and $Z_B = V_B^2/S_B$.
3. Divide each quantity by its base: line V by V_B, phase V by $V_B/\sqrt{3}$, Ω by Z_B, W, VAr and VA by S_B.
4. Solve in pu with no √3.
5. Multiply back: V(pu) × V_B gives a **line** voltage. I(pu) × I_B gives amps.

P25. X_S from OC/SC tests (Q4a)

1. Take both readings at the same I_F (with slopes, use $I_F = 1$ A).
2. $E_{0,ph} = V_{OC,L}/\sqrt{3}$, then $|Z_S| = E_{0,ph}/I_{SC}$.
3. $X_S = \sqrt{|Z_S|^2 - R^2}$ in Ω. Stop here if ohms are asked for.
4. $X_S(\text{pu}) = X_S/Z_B$. Sanity check: about 1–2.5 pu.

Check: Jun23 4a 3.676 Ω. Jun24 4a 2.137 pu (1.047 Ω).

P26. Sync machine phasor → E_0, δ, P_{max} (Q4a(ii), Q4b)

1. Everything in pu on the machine rating. $V = V\angle 0°$.
2. S = P/pf, $\sqrt{P^2+Q^2}$, or as given (“rated VA” = 1). $|I| = S/V$.
3. $\phi = -\cos^{-1}$pf for lag and $+$ for lead. Use the output current for a generator and the input current for a motor.
4. Generator: $E_0 = V + I(R + jX_S)$. Motor: $E_0 = V - I_m(R + jX_S)$. Work in rectangular form.
5. $|E_0|$ = excitation (pu), and $\delta = \angle E_0$ (+ for a generator, − for a motor).
6. If asked: $I_F = |E_0| \times I_F(1\text{ pu})$, $E_{0,L} = |E_0|V_B$, and $P_{max} = VE_0/X_S$ at 90°.

Check: Jun24 4b 1.634 pu, 82.43°, 0.9079 pu at 90°.

P27. PFC with a synchronous motor (Q4b, Jun23)

1. Each induction motor or load: $P_E = P_M/\eta$, then $Q = P_E\tan(\cos^{-1}\text{pf})$. This is positive (lag).
2. Sync motor: $P_E = P_M/\eta$, then $Q = -\sqrt{S_{rated}^2 - P_E^2}$ (fully over-excited, so leading).
3. $P_{tot} = \sum P$ and $Q_{tot} = \sum Q$.
4. pf $= \cos(\tan^{-1}(Q_{tot}/P_{tot}))$. Write **lag** if $Q_{tot} > 0$ and **lead** if it is negative.
5. If the motor’s excitation is asked for: $S_m = (P_E + jQ)/S_{rated}$ pu, $I_m = (S_m/V)^*$, then $E_0 = V - jX_SI_m$.

Check: Jun23 4b 235.3 kW, 158.1 kVAr; 222.2 kW, 114.5 kVAr; 0.9955 lag.

P28. Short answers, words only (Q4c, quiz/test last question)

1. No sketches or equations. Write quantities in words (“the rotor resistance divided by the slip”).
2. 3–5 bullets per part, each “fact … because …”. About 1 mark per point; about 1.5 min per mark.
3. Cover: what it is, the physical cause, the circuit element that models it, how it varies (with load, V, f, speed), and the practical consequence.

Timing: 30 min per question. Q1: (a) 5 min, (b) 6 min, (c)/(d) 15 min. Q2: (a) 12 min, (b) 7 min, (c) 9 min. Q3: 3a about 8 min, 3b about 7 min, 3c about 14 min; write the header block first. Q4: 4a X_S and phasors; 4b phasor/P_{max} or PFC; 4c words only.

Which method when

You see	Method	Where
Q1 “rated speed … torque … efficiency”, input V, I, P	$\omega = 2\pi n/60$, $\tau\omega$, η, $P/(\sqrt{3}V_L I_L)$	P1
Battery, open-circuit voltage, “voltage falls to”	$E = V_{OC}$, $R_S = (E - V)/I$	P2
“Three identical R on a V supply, star then delta”	V_1, I_1, I_L, $3I^2R$	P3
Series or parallel RLC with one phasor known; “S = VI*”	phasor Ohm’s law, $S = VI^*$	P4
“Draws I, absorbs P W and Q VAr; find R, X, L or C”	series/parallel formulas	P5
“Capacitor across the supply”, target pf, “find C”	PFC: P fixed, $Q_C = Q_S - Q_{old}$	P6
Peak, rms, period, phase delay, line/phase, turns ratio blanks	conversions	P7
“Sketch the power triangle”	P across, Q up (+) or down (–), S, $-\phi$	Ch2
Q2 Lengths, areas, μ_r, gap; “reluctance, inductance”	$\mathcal{R} = l/\mu A$, $L = N^2/\mathcal{R}$	P8
“DC current for B”, “energy in the gap”, “force”, “hold the core”	$NI = \sum Hl$, $B^2/2\mu_0$	P9
V, f and $\hat{B}$ or N; “use the back-emf equation”, “max voltage”	$E = 4.44Nf\hat{B}A$	P10
Iron loss with % hysteresis and eddy, new B and f	$B^2 f$, $B^2 f^2$ separately	P11
“Ideal transformer a:1”, impedances both sides	$Z' = n^2 Z$	P12

You see	Method	Where
OC and SC test readings	shunt from OC, series from SC	P13
“Delivering … at pf … at rated output voltage”	V_2', I_2', $V_1 = V_2' + I_2'(R + jX)$	P14
“Expression using KVL / voltage divider”, “lead or lag?”	symbolic KVL/divider; lags	P14
Copper loss, iron loss, efficiency; “load for max η”	$3I_2'^2R$, $3V_1^2/R_C$; $x = \sqrt{P_{FE}/P_{CU,FL}}$	P15
“Open circuit … input current and power”, “voltage halved”	$R_C \parallel jX_M$; ×0.5, ×0.25, ×0.25, ×1	P16
Q3 f and a rated speed; “poles? synchronous or induction?”	$p = 60f/n_R$, round down if non-integer	P17
“ideal / maximum possible efficiency”	$1 - s = n/n_S$ (DC: n/n_{NL})	P17
“fraction of rated torque”, “linear τ–n”	$n = n_0 - x\Delta n_R$; $P/P_R = x\,n/n_R$	P17
Name or sketch τ–n curves; “halve speed safely”	DC: V. AC: V and f	P18
DC k, R, V; no-load or stall data	$E = k\omega$, $\tau = kI$, $V = E + IR$	P19
Phase sequence, swapping leads, field magnitude	rotating-field rules	Ch6
No-load, blocked-rotor and DC test readings	per phase first	P20
“maximum torque”, “pull-out”, “speed at τ_{max}”	$s_{\tau\max} = R_2'/\lvert R_1 + jX\rvert$	P22
“external resistance”, “wound rotor”, “τ_{max} at standstill”	$R_{2EXT}' = \lvert R_1 + jX\rvert - R_2'$	P22

You see	Method	Where
Speed above n_S, "run as generator"	full solve with $s < 0$, keep the signs	P21
"use the torque formula", only τ asked	torque formula with phase V_1, ω_S	Ch7
I_1, pf, P_{IN}, P_{AG}, η asked	full equivalent-circuit solve	P21
"inverter", "volts per hertz"	scale V_1, X, ω_S with f	Ch7
Q4 Excitation / OC voltage / SC current table, or "V per field amp"	X_S from tests, then Ω or pu as asked	P25
"delivering P MW at pf ...", "x % above/below rated"	per-unit V and I phasors	P24, P26
"excitation required", "rotor angle"	E_0, δ	P26
"excitation fixed ... maximum power without losing synchronism"	$P_{max} = VE_0/X_S$ at 90° (state δ = 90°)	P26
Plant of induction motors plus a sync motor (kVA, η)	PFC with $\sqrt{S^2 - P_E^2}$	P27
"words only" (Q4c, quiz Q7, test Q8)	short-answer points	P28
Generation, transmission, renewables	Ch9 table and PQ9.1	Ch9

Traps

Ch1

> rpm in $P = \tau\omega$. Convert with $2\pi/60$ first.

> Treating the motor rating as input power. It is the shaft output: $P_{IN} = P_{rated}/\eta$.

> Using $\sqrt{3}$ with phase quantities or 3 with line quantities.

Quoting pf without “lead” or “lag”.

Peak vs rms: the 4.44 equation and every power formula use rms V and I (peak B).

Ch2

Forgetting the conjugate in $S = VI^*$ (flips the sign of Q).

Adding magnitudes ($50 + 20$ V) instead of phasors ($\sqrt{50^2 + 20^2}$).

Adding apparent powers of loads. Add P’s and Q’s.

Taking the sign of Q from the square root. L or lag is +, C or lead is −.

Using V^2/P (parallel) when the question wants series R, or the reverse.

Battery load power as EI or loss as V^2/R_S. They are VI and I^2R_S.

Putting V_L across a star phase. The supply figure is always line.

PFC: using V_L for star capacitors, using 50 Hz when the supply is 60 Hz, giving Q_C as positive, or thinking P changes.

Atan quadrant: if Re < 0, add 180°.

Ch3

cm² → m² is 10^{-4}; mm → m is 10^{-3}; radius vs diameter.

Missing the second gap or the second pole face (2 × 0.07 mm, 10 cm²).

Putting reluctance or μ_r in the back-emf equation.

Rounding N down (B then exceeds the limit).

Scaling the total iron loss with one law. Split into hysteresis and eddy first.

$Z' = nZ$ instead of n^2Z; referring in the wrong direction.

Series formulas on the OC test (gives a series R, X, not $R_C \parallel X_M$).

Forgetting per phase on 3ph tests ($V_L/\sqrt{3}$, $P/3$).

P_{CU} with I_1 or P_{FE} with V_2'; forgetting the ×3 for 3ph losses.

Using V_1's line value in the per-phase KVL, or forgetting $\sqrt{3}$ when the line voltage is asked.

Lead sign: a load **generating** Q has a leading current, $+\phi$.

Saying V_2' is in phase with V_1 for a resistive load. It lags.

Ch4

Taking p as poles in $n_S = 60f/p$. 6-pole at 50 Hz is 1000 rpm, not 500.

Rounding p to the nearest integer instead of down, for an induction machine.

Taking part-load power as xP_R. It is $x\,n/n_R$.

Scaling the rated speed in proportion. The speed drop (DC) or slip speed (IM) stays constant.

Changing f without V on an AC machine (saturation). “Safe” means V and f together.

Ch5

Using rpm in $E = k\omega$.

Taking $P_{IN} = EI$. It is VI.

Using E/V for generator efficiency. It is V/E, with $E = V + |I|R$.

Ch6

Thinking a cyclic relabelling reverses rotation. Only swapping two leads does.

Saying a single-phase field rotates. It pulsates.

Ch7

Putting the line voltage into the per-phase circuit. That makes every power ×3.

Dividing P_{AG} by ω. Use ω_S; only P_M goes with ω.

Taking rotor Cu as $3I_2'^2R_2'/s$. That is P_{AG}; rotor Cu is sP_{AG}.

Adding $I_2' + I_M$ as magnitudes.

Taking the slip at the rated speed when another operating speed is given.

Rounding s early (0.02 vs 0.02222 is 11 % in R_2'/s).

Using line values in the tests. Apply $V_L/\sqrt{3}$ and $P/3$ first. $X = Q_1/I_1^2$, not V/I.

Using $R_1 + X$ in $s_{\tau\max}$. Thinking extra rotor R raises τ_{max} (it moves it).

Dropping the minus signs in generator mode. Angles near ±180° are correct.

V/f: scaling only V. Also scale X and ω_S.

Ch8

Using the OC line voltage as a phase voltage. That makes X_S √3 too big.

Answering in Ω when pu is asked, or the reverse (Jun23 asks Ω, Jun24 asks pu).

Computing Z_B with phase V and total VA.

Treating MW as MVA. $S = P/\text{pf}$.

Using V = 1 when the question says ±x % of rated.

Wrong sign on the current angle. Lead is $+\cos^{-1}$pf; a wrong sign gives 2.41 pu instead of 1.634 in Jun24 4b.

Using $E_0 = V + jX_SI$ with a motor's **input** current.

Recomputing E_0 for P_{max}. Also forgetting to state δ = 90°.

> Using P_M instead of $P_E = P_M/\eta$ in the capability triangle.

> Leaving “lag” or “lead” off the final pf.

Ch9

> Saying high voltage “increases power”. It reduces current, so I^2R loss (and conductor size) for the same power.

> Mixing up the voltage levels: generation 10–25 kV, transmission 115–800 kV, distribution 2.4–69 kV, domestic 240 V.

> Listing only one renewable difficulty. Give economic, technical and environmental.

Past-paper anchors

All recomputed in Python. Values in brackets are the official sheet where it differs (rounding unless noted).

Paper	Result
Jun23 1a	E = 13.8 V; $R_S = 0.01433\ \Omega$, 1290 W, 2850 W; 13.51 V
Jun23 1b	star 3.794 A, 3.794 A, 3.023 kW; delta 6.571 A, 11.38 A, 9.069 kW
Jun23 1c	$V_R = 130\angle 20^\circ$, $V_L = 390\angle 110^\circ$, $V_C = 650\angle -70^\circ$ V; $Z = 22.36\angle -63.43^\circ\ \Omega$, $V_1 = 290.7\angle -43.43^\circ$ V; $I_1^* = 13\angle -20^\circ$, S = 1690 − j3380 VA (−j3379)
Jun23 2a	596.8 kA/Wb, 66.31 kA/Wb, 1.544 mH; 3.867 J; 3.84 W, 14.93 W, 18.77 W
Jun23 2b	R = 0.5845 Ω, X = 2.721 Ω, $R_C = 1280\ \Omega$, $X_M = 186.6\ \Omega$
Jun23 2c	KVL and divider expressions; V_2' lags (series X)
Jun23 3a	10 poles, induction, η = 93.33 %, 590 rpm, 26.34 %
Jun23 3b	$R'_{2EXT} = 2.328\ \Omega$
Jun23 3c	4 poles, 1500 rpm; generator at 1520 rpm; τ = 101.7 Nm; $I_1 = 29.79\angle -38.44^\circ$ A
Jun23 4a	$X_S = 3.676\ \Omega$; $V = 1\angle 0^\circ$, $I = 0.7454\angle -26.57^\circ$ pu (0.7453)
Jun23 4b	IM 235.3 kW, 158.1 kVAr; sync 222.2 kW, 114.5 kVAr (114.6); pf 0.9955 lag
Jun24 1a	n = 10, step down; n = 0.5, step up; 700 W, 70 %, decrease, increase
Jun24 1b	pf 0.8184 lag; 3.420 kW, 85.50 %
Jun24 1c	R: 400 W, 0 VAr; L: 0 W, 160 VAr; $V_1 = 53.85$ V
Jun24 1d	pf 0.8 lag; $P_S = 80$ W, $Q_S = 38.75$ VAr, $Q_C = -21.25$ VAr; C = 6.017 µF

Paper	Result
Jun24 2a	810.6 kA/Wb, 17.77 mH; 19.62 V; 439.5 N
Jun24 2b	0.3, 10.8, 25.8 Ω; 0.4651 A, 2.791 A, 0.8372 V, 5.023 V
Jun24 2c	0.14 Ω, j0.28 Ω, 250 Ω, j14 Ω; 660.9 A, 1.024 MW; ×0.5, ×0.25, ×0.25, ×1
Jun24 3b	$R_1 = 2\ \Omega$, $R_2' = 3.2\ \Omega$, $X = 7.635\ \Omega$
Jun24 3c	1000 rpm, s = −0.025, $R_2'/s = -120\ \Omega$; $I_1 = 2.029\angle -174.2^\circ$ A, $P_{IN} = -1451$ W; $P_{AG} = -1481$ W (−1.482 kW), τ = −14.15 Nm
Jun24 4a	$X_S = 2.137$ pu (1.047 Ω); $V = 0.97\angle 0^\circ$, $I = 0.6443\angle -36.87^\circ$ pu
Jun24 4b	$I = 1\angle +25.84^\circ$; $E_0 = 1.634$ pu, δ = 82.43° (82.42°); $P_{max} = 0.9079$ pu (0.9077) at δ = 90°
2023 quiz Q1	1.1 kW, 1.341 kW, 0.241 kW; 7.346 Nm (149.7 rad/s)
2023 quiz Q2	R = 49.31 Ω, X = 0.9028 Ω, inductive, L = 2.874 mH; $V_L = 30.93\angle -5.96^\circ$ V
2023 quiz Q3	N = 360 (or 361); L = 0.4 H, X = 125.7 Ω (125.6); I = 0.7958 A (0.7955)
2023 quiz Q4	I = 46.69 A; F = 962.9 N
2023 quiz Q5	1.179 kW, 1.369 kW, η = 97.78 % (97.79)
2023 quiz Q6	$V_2' = 1.905\angle 0^\circ$ kV, $I_2' = 6.056\angle -38.74^\circ$ A; $V_1 = 2.254\angle 5.89^\circ$ kV, line **3.904 kV** (sheet 3.883 kV, inconsistent with its own 2.254 kV)
2024 test Q1	169.7 V, 120 V, 16.67 ms, 4.167 ms; 400 V, 230.9 V, 15 A, 15 A
2024 test Q2	$S = 1200 + j400 = 1265\angle 18.43^\circ$ VA, 1265 VA; pf 0.9487 lag

Paper	Result
2024 test Q3	$V_1 = 240\angle 0^\circ$, $I_R = 2\angle 0^\circ$, $I_L = 4\angle -90^\circ$, $I_1 = 4.472\angle -63.43^\circ$ A; 480 W, 960 VAr
2024 test Q4	0.3519 T; 0.1086 T; 0.0860 T
2024 test Q5	169 turns; 3.453 Ω; 34 turns
2024 test Q6	X = 1.005 Ω, R = **0.3068 Ω** (sheet 0.3086, digit swap: 46 667/390² = 0.3068)
2024 test Q7	$V_2' = 8.083\angle 0^\circ$ kV, $I_2' = 6.186\angle +13.49^\circ$ kA

Short-answer points

Words only. About one mark per point; give 3–5 per part.

Ch1–2 (Q1, and Q4c has asked these)

- *Real, reactive, apparent power (Jun24 4c(i)):* P comes from the current in phase with V; it is the average of instantaneous power, net energy to the load, does useful work, absorbed by resistors. Q comes from the quadrature current; energy oscillates between source and load with zero mean; inductors absorb it, capacitors generate it; no useful work but needs current. S is the magnitude of complex power (real part P, imaginary part Q), equals V × total I, and sets ratings. P and Q are both scalars.
- *PFC (Jun23 4c(i)):* loads (mainly induction motors) draw Q, which is energy oscillating in the system. It loads generators, transformers and lines and adds losses. A parallel capacitor (or over-excited sync motor) generates Q locally, so the supply Q, S and current fall. P and the load are unchanged. Tariffs on VA make it pay.
- *Why three-phase:* a balanced 3ph winding makes a constant rotating field, so machines self-start; total instantaneous power is constant (smooth, quiet); better use of material and conductors.
- *Why efficiency matters:* limited stored energy (EVs), running cost over the life, and heat removal limits the rating.

Ch3 (Q2, quiz and test last questions)

- *Choosing primary turns (2023 quiz Q7):* the back-emf equation links turns and peak flux for a given V and f; choose turns so B is just below saturation. Too few: B too high, saturation, high and distorted current, overheating. Too many: low B, more winding R, more copper loss, lower η, possible overheating.
- *Ideal vs real, load R doubled (2024 test Q8):* ideal: V set only by input V and turns ratio, so unchanged; current halves (Ohm's law). Real: series R and leakage X drop less voltage at lower current, so V rises slightly; current falls to slightly more than half.

- *Transformer losses (Jun24 4c(ii)):* copper loss: I^2R heating of the windings, series winding R, varies with the square of output. Iron loss: hysteresis (domain reversal) and eddy currents (induced core currents), core-loss resistance in the shunt branch, constant with load (set by V and f). Max η where they are equal.
- V_2' *phase, resistive load (Jun23 2c(iii)):* lags, because the series reactance makes the current lag V_1 and the load voltage is in phase with the current.
- *OC voltage halved (Jun24 2c(iii)):* fixed shunt impedance: current ×0.5, P and Q ×0.25, pf ×1. A real core draws even less current (further below the knee).
- *Transformer no-load current not zero:* magnetising current (finite permeability, X_M, lags 90°) plus core-loss current (hysteresis and eddy, R_C, in phase). Saturation makes it large and distorted at high V.
- *Airgap effect:* μr = 1, so even a short gap dominates the reluctance; L falls, more current for the same flux, but L becomes linear; most energy and the force are in the gap; in an IM the gap is unavoidable, so it is kept small (magnetising current, pf).
- *Laminations and silicon steel:* thin insulated laminations break eddy-current paths; silicon raises resistivity and lowers hysteresis loss.

Ch7 (Q4c has asked these)

- *IM no-load current not zero (Jun24 4c(iii)):* magnetising current sets up the airgap flux, and is large because of the airgap (X_M). A core-loss component covers hysteresis and eddy currents (R_C). Friction (bearings) and windage (air, fan) need a little power, so there is a small slip and rotor current. The pf is low.
- *Perfectly conducting rotor,* $R_2' = 0$ *(Jun23 4c(iii)):* no rotor loss, so higher efficiency. $s_{\tau\max} = 0$, so the curve is vertical at n_S, like a synchronous machine. Zero starting torque.
- *Transformer vs IM (Jun23 4c(ii)):* both have primary and secondary windings on a magnetic core, both run on 3ph AC, with similar circuits. The IM secondary rotates and is short-circuited, the rotor frequency is sf, the rotor R is divided by s, and the airgap makes the magnetising current much larger.
- *Torque production:* rotating field at n_S; slip induces slip-frequency emf in the shorted rotor; rotor currents in the field give force; at n_S no change of flux, so no emf, current or torque.
- *V/f control:* frequency sets n_S; flux $\propto V/f$; reducing f alone saturates the core (large current); keeping V/f constant keeps flux and τ_{max}, so the curve slides along the speed axis; small boost at low f for the R_1 drop.
- *Tests:* DC test gives R_1; no-load (s ≈ 0) gives R_C, X_M (includes friction and windage); blocked rotor (s = 1, low V) gives $R_1 + R_2'$ and X; split X equally.
- *Wound rotor and external R:* raises $s_{\tau\max}$ and starting torque, lowers starting current, τ_{max} unchanged; extra rotor loss, so cut it out at speed.
- *IM as generator:* driven above n_S, s < 0, R_2'/s negative, power reverses; still absorbs Q for magnetising; cannot control its own Q or voltage; suits wind and regenerative braking.

Ch8

- *Sync machine advantages:* exact speed, high efficiency, and controllable Q (unlike an IM).
- *Induction vs synchronous:* sync runs exactly at n_S, IM below it with slip rising with load; sync rotor field from DC or magnets, IM rotor currents induced; IM always absorbs Q

(lagging), sync Q set by field current; IM self-starting, cheap, robust.

- *Industrial pf lagging, why sync motors:* plants are mostly induction motors, which absorb Q; VA tariffs and cable loading; an over-excited sync motor generates Q while driving its load; limited to $\sqrt{S^2 - P^2}$.
- *Field current on an infinite bus:* V and f fixed by the system; P set by the turbine; I_F sets E_0 and Q; over-excited delivers Q (lagging), under-excited absorbs Q (leading, larger δ, less stable); V-curve minimum at unity pf.
- *Q capability limit:* stator current heating (S_{rated}, so $Q = \sqrt{S^2 - P^2}$), rotor field heating on the over-excited side, stability (δ → 90°) on the under-excited side.
- *Per-unit advantages:* values relative to rating show size at a glance; similar machines have similar pu values (easy error check); no √3 or 3, line = phase; transformer impedances the same on both sides.
- *Salient vs round rotor:* low speed (hydro) needs many poles, easy with salient poles, low stress; high speed (2–4 poles, 3000 rpm) needs a strong round rotor with slotted winding, low windage.
- *Not self-starting:* at standstill the field sweeps past and the torque alternates each half cycle, average zero; start with a pony motor, damper (starting) cage or an inverter ramp.
- *Sync vs DC circuit:* both an emf proportional to speed and field behind a series impedance; DC uses R and DC quantities; sync uses mainly X_S and phasors, where δ controls P and $|E_0|$ controls Q.

Ch9

- *High-voltage transmission:* same power, current $\propto 1/V$; loss $\propto I^2$, so loss falls with V^2; thinner conductors, less voltage drop; transformers make stepping cheap.
- *Paralleled lines and generators:* reliability (fault tolerance), maintenance without interrupting supply, run the most efficient number of generators.
- *Renewables:* unlimited resource, zero operating emissions; but economic (cost, subsidies), technical (variability, storms) and environmental (visual, noise) difficulties.
- *Fossil fuels:* finite reserves; CO_2 (global warming), SO_x and NO_x (acid rain); nuclear has radioactive waste.